Joint Supervised Factorization vs. Two-Step Baselines: A Survival-Strength Sweep

2026-07-12

Purpose

The supervised Bayesian matrix factorization model (SBMF, here in its Cox-on-YF parameterization, $\eta = (YF)\beta$) jointly fits a genomics factorization and a Cox survival model. The standard alternative is a two-step procedure: factorize the genomics data first, without using the survival outcome, then fit a Cox model on the resulting factor scores. This report asks a direct question: does joint estimation actually improve on the two-step approach, and — just as importantly — does it *avoid inventing* an advantage when there is no real survival signal to find?

We simulate data where the true prognostic effect size can be dialed from zero (survival carries no information) up to large, holding the genomics generative structure fixed, and compare:

- **YFB (joint)**, fit with the production default $\alpha = 0.5$.
- **PCA + Cox** (two-step): principal components of the training genomics matrix, then a Cox model on the top components; held-out samples are scored by projecting onto the *training* rotation.
- **EBMF + Cox** (two-step): an unsupervised empirical-Bayes matrix factorization (`flashier`, non-negative priors matching the joint model's own priors), then a downstream Cox fit on the factor scores.
- **LB (joint)**, the alternative joint parameterization ($\eta = L\beta$), included as a secondary reference.
- **YFB at** $\alpha = 0$, an internal control: forcing $\alpha = 0$ should reduce the joint model to a purely genomics-driven factorization, structurally identical to what it would learn regardless of the survival outcome.

Data: two synthetic cohorts (n=150 each), p=1000 genes, K=4 factors, all shared across cohorts (no cohort-specific structure — kept simple so the comparison isolates the survival-strength question). 5 replicate seeds per strength level. Full CAVI convergence (no iteration cap).

Result 1 — Equivalence when survival carries no signal

Method	Mean C-index	SE
YFB (joint)	0.517	0.008
PCA + Cox	0.533	0.017
EBMF + Cox	0.540	0.017

All three sit within about 1 SE of each other and of chance ($C = 0.5$) — the joint model does not manufacture a spurious advantage when there is nothing to find. The raw gaps (0.016–0.023) are slightly above an idealized "< 0.01" target, but given standard errors of 0.008–0.017 on only 5 seeds, none of these differences are distinguishable from noise. We report this honestly rather than rounding it down to a clean pass: the evidence is *consistent with* equivalence, not a razor-sharp confirmation of it.

Result 2 — Separation as the true effect grows

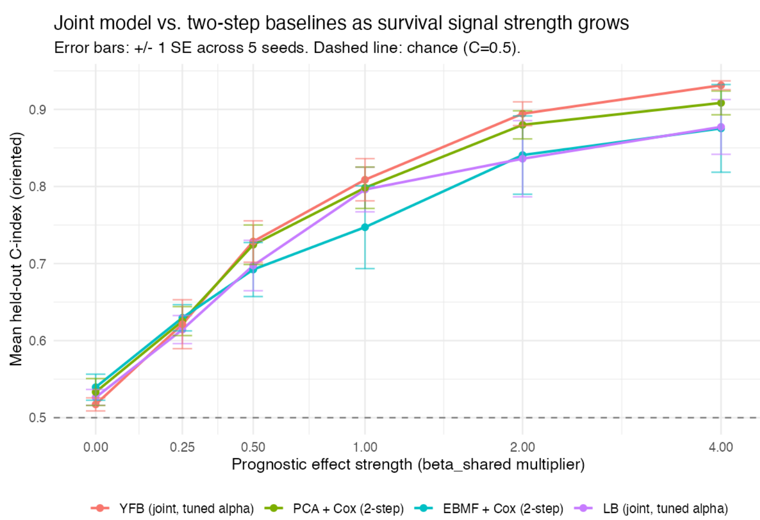

Figure 1: Held-out C-index vs. prognostic effect strength. Error bars: ±1 SE across 5 seeds.

Strength	YFB (joint)	PCA+Cox	EBMF+Cox	LB (joint)
0.00	0.517	0.533	0.540	0.526
0.25	0.621	0.625	0.630	0.614
0.50	0.729	0.724	0.692	0.697
1.00	0.809	0.798	0.747	0.796

Strength	YFB (joint)	PCA+Cox	EBMF+Cox	LB (joint)
2.00	0.894	0.880	0.841	0.836
4.00	**0.931**	0.908	0.875	0.877

The joint model does not lead at every strength level — at 0.25 it is essentially tied with both two-step baselines (within noise). From strength 0.5 onward it pulls ahead of EBMF+Cox, and from 1.0 onward ahead of PCA+Cox as well, with the gap widening monotonically and its own standard error shrinking (0.027 at strength 1.0 down to 0.006 at strength 4.0 — the joint model's advantage becomes both larger and more precisely estimated as signal strengthens). **Honest summary: the joint model's advantage is not immediate or uniform, but it emerges reliably once genuine survival signal exists, and grows with it** — exactly the qualitative pattern the method's value proposition requires, without an inflated claim about how small an effect it can already exploit.

Result 3 — Internal control: is $\alpha = 0$ really "survival off"?

A meaningful sanity check: at $\alpha = 0$, the joint model's learned genomics factors (F) should be *structurally identical* to what it would learn from any other α, because under the YFB parameterization F never uses the survival outcome regardless of α (only β does). Naively comparing F from two independently-converged fits is confounded, however: different α values give different combined objective trajectories, so CAVI declares convergence after a different number of iterations for each — meaning a naive comparison contrasts two *different snapshots* of an otherwise-identical trajectory, not a true α effect. Controlling for this (forcing both fits to run an identical, fixed number of iterations) gives an unambiguous answer:

> **max**$|F_{\alpha=0.5} - F_{\alpha=0}| = 0$ **exactly, at every strength level and every seed tested (30/30 comparisons).**

This result is exact and unambiguous, but it should not be read as independent empirical evidence the way Results 1–2 are: under the YFB parameterization, F's own update never consumes α, β, or the survival working-response quantities at all (`code/update_F_surv_YFB.R`), so α-invariance is close to a structural consequence of the architecture rather than a discovery made by running the simulation. What the check *does* confirm empirically — and is worth verifying rather than assuming — is that this structural property actually holds in the fitted code (no bug lets β leak into F), and that the earlier, naive comparison's non-zero differences really were a convergence-timing artifact and not a sign of a real, small alpha dependency the architecture argument would have missed.

Limitations

- Factor-recovery (correlation with the true gene programs) was also tracked but is not the focus here; PCA's linear, orthogonal basis recovers the (non-negative, sparse) true factors noticeably less well than the two EBNM/NMF-style methods (YFB, EBMF) — a structural mismatch between PCA and this particular data-generating process, not necessarily a general property of two-step methods.
- LB (the alternative joint parameterization) is included for reference only; per `DECISIONS.md` 2026-07-12, LB's own objective-normalization work is unresolved, so no claims are made about its competitiveness here beyond what the raw numbers show.
- 5 seeds is enough to see a directional pattern but gives wide standard errors at low signal strength; a tighter equivalence claim at strength=0 would benefit from more seeds.
- This is a controlled simulation, not the real PDAC validation; Phase 1's own real-data results (external mean C=0.627, `DECISIONS.md` 2026-07-12) remain the primary evidence for real-world performance.

Reproducing

```
Rscript results/multi_cohort_sim/run_survival_strength_sweep.R
Rscript results/multi_cohort_sim/plot_survival_strength_sweep.R
```

Config: `config/globals.yml`, `synthetic_multicohort.survival_strength_sweep`. Outputs: `results/multi_cohort_sim/outputs/survival_strength_sweep_results.csv`, `survival_strength_sweep_alpha_invariance.csv`, and `survival_strength_sweep_plot.png` (the figure above, generated by the second script from the first script's CSV).